

Practical No. 8

Introduction to MySQL

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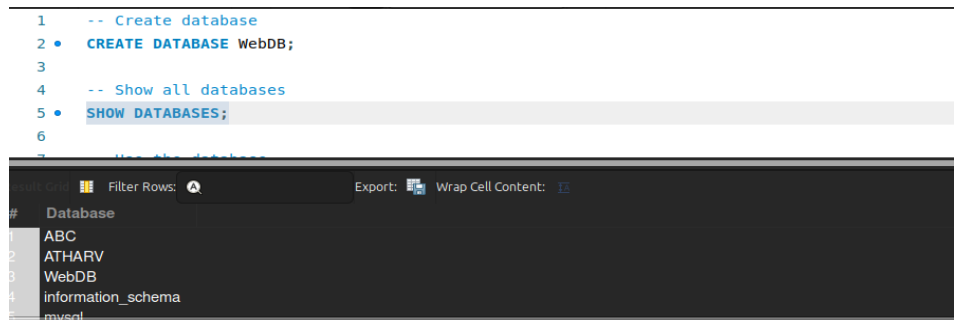
Batch :CSE-S5

Problem Statement 1: Installing and Exploring MySQL

- Install MySQL on your system.
- Open the MySQL command-line client or GUI tool (e.g., MySQL Workbench).
- Create a new database named **WebDB**.

```
1  -- Create database
2 • CREATE DATABASE WebDB;
3
4  -- Show all databases
5 • SHOW DATABASES;
6
7  -- Use the database
8 • USE WebDB;
```

- List all databases available on your system.



Problem Statement 2: Creating a Simple Table

- Create a table **Users** with the following fields:
 - o **UserID** (INT, Primary Key)
 - o **Username** (VARCHAR(30))
 - o **Email** (VARCHAR(50))

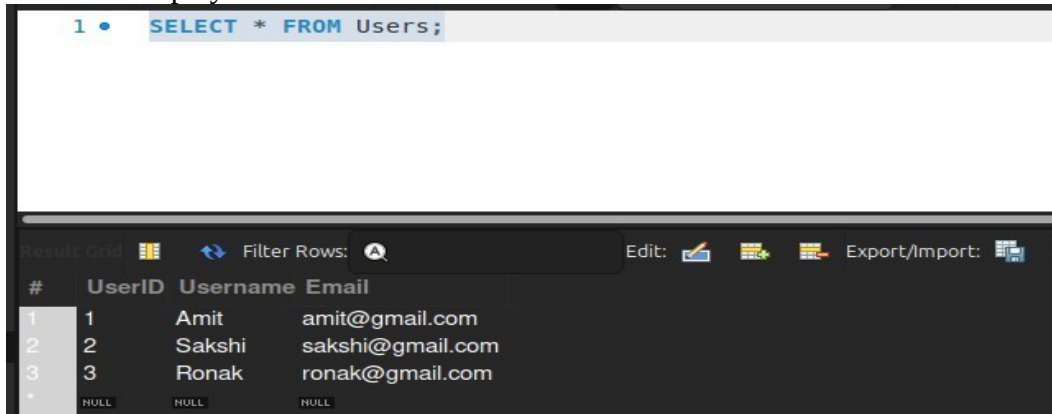
```
1 • CREATE TABLE Users (  
2     UserID INT PRIMARY KEY,  
3     Username VARCHAR(30),  
4     Email VARCHAR(50)  
5 );
```

- Insert at least 3 sample records into the table.

```
1 • INSERT INTO Users VALUES (1, 'Amit', 'amit@gmail.com');  
2 INSERT INTO Users VALUES (2, 'Sakshi', 'sakshi@gmail.com');  
3 INSERT INTO Users VALUES (3, 'Ronak', 'ronak@gmail.com');
```

Problem Statement 3: Basic Data Retrieval

- Write queries to:
 - o Display all records from the **Users** table.

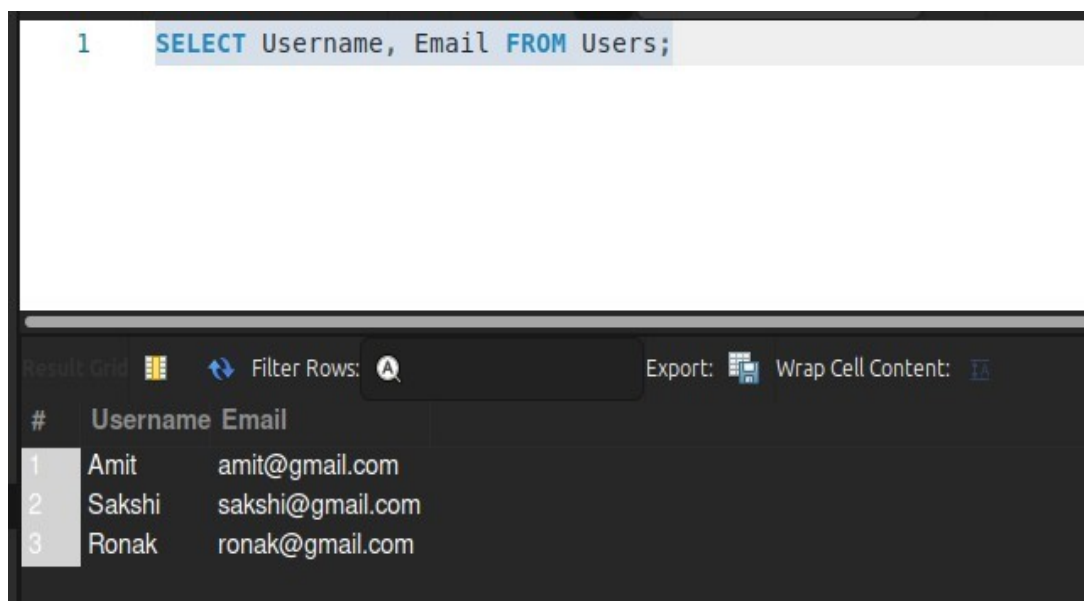


```
1 • SELECT * FROM Users;
```

Result Grid

#	UserID	Username	Email
1	1	Amit	amit@gmail.com
2	2	Sakshi	sakshi@gmail.com
3	3	Ronak	ronak@gmail.com
*	NULL	NULL	NULL

- o Retrieve only the **Username** and **Email** fields.

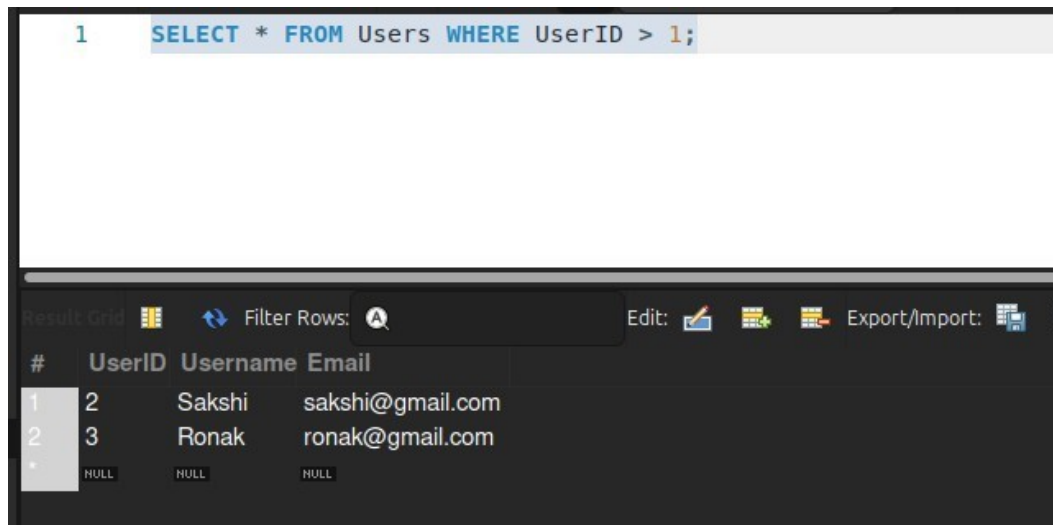


```
1 SELECT Username, Email FROM Users;
```

Result Grid

#	Username	Email
1	Amit	amit@gmail.com
2	Sakshi	sakshi@gmail.com
3	Ronak	ronak@gmail.com

- o Display records where **UserID** is greater than 1.



The screenshot shows a database query interface. At the top, a SQL query is entered in a text box: `1 SELECT * FROM Users WHERE UserID > 1;`. Below the query box, there is a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. The 'Result Grid' is active, displaying a table with the following data:

#	UserID	Username	Email
1	2	Sakshi	sakshi@gmail.com
2	3	Ronak	ronak@gmail.com
*	NULL	NULL	NULL

Problem Statement 4: Updating and Deleting Records

- Write queries to:
 - o Update the email of a specific user.

```
1  UPDATE Users
2  SET Email = 'amit_new@gmail.com'
3  WHERE UserID = 1;
4  • SELECT * FROM Users;
```

Result Grid

#	UserID	Username	Email
1	1	Amit	amit_new@gmail.com
2	2	Sakshi	sakshi@gmail.com
3	3	Ronak	ronak@gmail.com

- o Delete a user record based on **UserID**.

```
1  • DELETE FROM Users
2  WHERE UserID = 3;
3  • SELECT * FROM Users;
```

Result Grid

#	UserID	Username	Email
1	1	Amit	amit_new@gmail.com
2	2	Sakshi	sakshi@gmail.com
*	NULL	NULL	NULL

Problem Statement 5: Integration with Web Development

- Write a simple PHP/Node.js script to connect to the **WebDB** database.

Connects PHP to MySQL database **WebDB** Retrieves data from Users table
Displays records on the web page Handles error if connection

```
ronakkumar@ronakkumar:~/Desktop$ cat index.php
<?php
// Database connection
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "WebDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection (Error handling)
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// SQL query
$sql = "SELECT * FROM Users";
$result = $conn->query($sql);

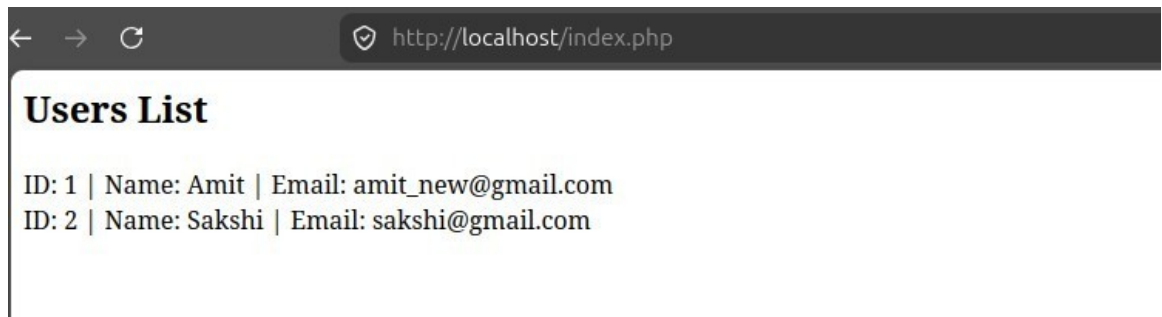
// Display data
if ($result->num_rows > 0) {
    echo "<h2>Users List</h2>";
    while($row = $result->fetch_assoc()) {
        echo "ID: " . $row["UserID"] .
            " Name: " . $row["Username"] .
            " Email: " . $row["Email"] . "<br>";
    }
} else {
    echo "No records found";
}

// Close connection
$conn->close();
?>
```

- Retrieve and display the records from the Users table on a web page.

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- Ensure proper error handling if the connection fail

